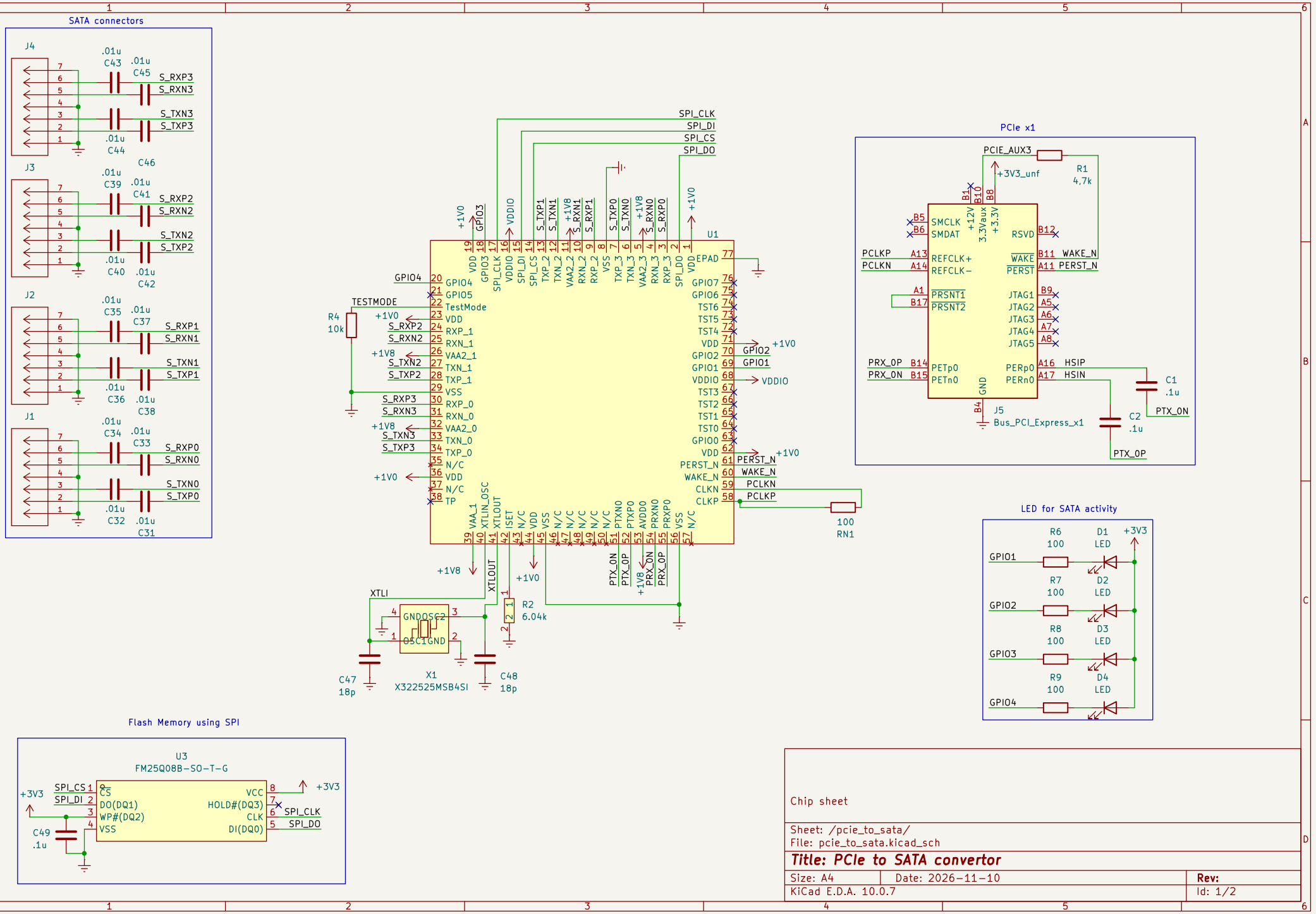
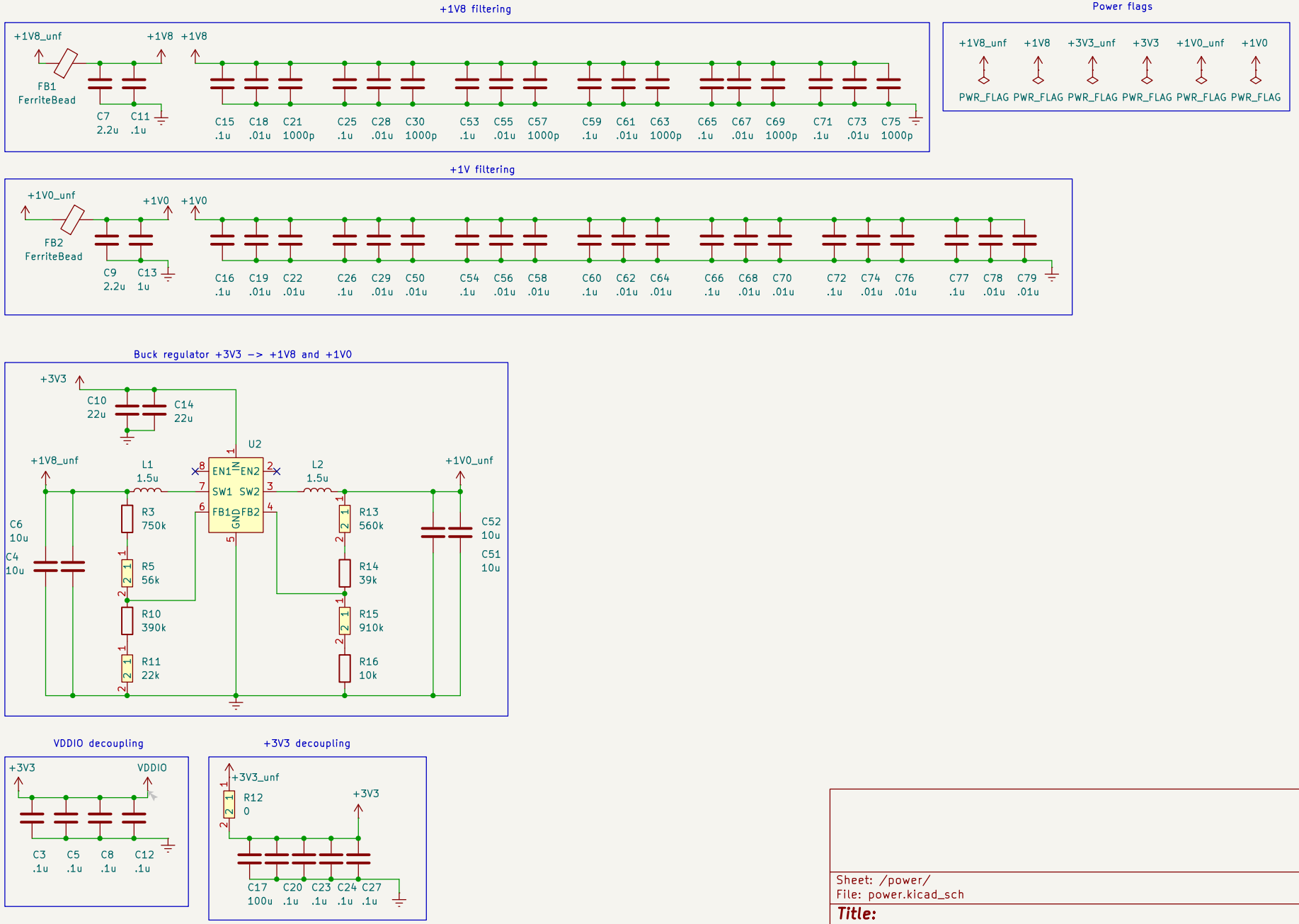

SATA connectors
J4
J3
J2
J1
S_RXP3
S_RXN3
S_TXN3
S_TXP3
S_RXP2
S_RXN2
S_TXN2
S_TXP2
S_RXP1
S_RXN1
S_TXN1
S_TXP1
S_RXP0
S_RXN0
S_TXN0
S_TXP0
U1
SPI_CLK
SPI_DI
SPI_CS
SPI_DO
TESTMODE
R4
10k
R2
6.04k
R1
4,7k
100
RN1
XTLI
XTLOUT
X1
X322525MSB4SI
C47
18p
C48
18p
PCIe x1
PCIE_AUX3
+3V3_unf
PCLKP
PCLKN
PRX_0P
PRX_0N
PTX_0N
PTX_0P
J5
Bus_PCI_Express_x1
C1
.1u
C2
.1u
LED for SATA activity
GPIO1
GPIO2
GPIO3
GPIO4
D1
D2
D3
D4
LED
Flash Memory using SPI
U3
FM25Q08B-SO-T-G
C49
.1u
Chip sheet
Sheet: /pcie_to_sata/
File: pcie_to_sata.kicad_sch
Title: PCIe to SATA convertor
Size: A4
Date: 2026-11-10
Rev:
KiCad E.D.A. 10.0.7
Id: 1/2


+1V8 filtering
+1V8_unf
+1V8
+1V8
FB1
FerriteBead
C7
2.2u
C11
.1u
C15
.1u
C18
.01u
C21
1000p
C25
.1u
C28
.01u
C30
1000p
C53
.1u
C55
.01u
C57
1000p
C59
.1u
C61
.01u
C63
1000p
C65
.1u
C67
.01u
C69
1000p
C71
.1u
C73
.01u
C75
1000p
Power flags
+1V8_unf
+1V8
+3V3_unf
+3V3
+1V0_unf
+1V0
PWR_FLAG
PWR_FLAG
PWR_FLAG
PWR_FLAG
PWR_FLAG
PWR_FLAG
+1V filtering
+1V0_unf
+1V0
+1V0
FB2
FerriteBead
C9
2.2u
C13
1u
C16
.1u
C19
.01u
C22
.01u
C26
.1u
C29
.01u
C50
.01u
C54
.1u
C56
.01u
C58
.01u
C60
.1u
C62
.01u
C64
.01u
C66
.1u
C68
.01u
C70
.01u
C72
.1u
C74
.01u
C76
.01u
C77
.1u
C78
.01u
C79
.01u
Buck regulator +3V3 -> +1V8 and +1V0
+3V3
C10
22u
C14
22u
U2
+1V8_unf
L1
1.5u
EN1
EN2
SW1
SW2
FB1
GND
FB2
L2
1.5u
+1V0_unf
R3
750k
R5
56k
R10
390k
R11
22k
R13
560k
R14
39k
R15
910k
R16
10k
C6
10u
C4
10u
C52
10u
C51
10u
VDDIO decoupling
+3V3
VDDIO
C3
.1u
C5
.1u
C8
.1u
C12
.1u
+3V3 decoupling
+3V3_unf
R12
0
+3V3
C17
100u
C20
.1u
C23
.1u
C24
.1u
C27
.1u
Sheet: /power/
File: power.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.7
Id: 2/2